

Kalai’s Conjecture for Tight Trees

Dhruv Mubayi*

Jacques Verstraëte†

Abstract

Let $r \geq 2$ and $t \geq 1$. It is shown that if T is an r -uniform tight tree with t edges and H is a T -free r -uniform hypergraph, then $|E(H)| \leq (t-1)|\partial H|/r$, where ∂H is the $(r-1)$ -shadow of H . The bound is tight infinitely often, and establishes Kalai’s Conjecture, whose $r = 2$ case is the Erdős-Sós Conjecture.

The proof was found by GPT-6 Astra, extending its method of proof for the Erdős-Sós conjecture to the hypergraph setting. It is noteworthy that previous proofs of special cases of the Erdős-Sós conjecture do not extend to give tight bounds in the hypergraph setting.

A strengthening of the Erdős-Sós conjecture due to the authors about tight lower bounds on the number of copies of a tree in a graph with average degree $d \geq t-1 \geq 0$ remains open.

1 AI Declaration Statement

The proof of the main result was found by GPT-6 Astra by first prompting it to extend its recent proof of the Erdős-Sós conjecture (see Adamczewski and Bloom [6, Appendix B.4]) to the case of hypergraph tight paths as in [14], and then prompting it to prove the full Kalai conjecture. As we elaborate throughout the text, GPT-6 Astra’s proof technique for Erdős-Sós conjecture, which uses permutations of the underlying vertex set, perhaps has its genesis in an old argument of Perles (see [19]), which considered convex geometric graphs (these corresponds to cyclic permutations of the vertex set). This method was developed in the hypergraph case by Füredi-Jiang-Kostochka-Mubayi-Verstraëte in [14] and gave the previously best bounds for Kalai’s conjecture for tight paths. The authors have checked the proof and rewritten much of the text provided by GPT-6 Astra and added further explanations throughout the paper. They take full responsibility for all the content in the paper.

2 Introduction

All hypergraphs in this paper are finite and simple. An r -uniform hypergraph, or r -graph, has edges of size r . For an r -graph T , let $\text{ex}(n, T)$ denote the maximum number of edges in an n -vertex r -graph containing no copy of T .

*Department of Mathematics, Statistics and Computer Science, University of Illinois, Chicago, IL 60607. Email: mubayi@uic.edu. Research partially supported by NSF Awards DMS-2552740 and DMS-2153576.

†Department of Mathematics, University of California, San Diego, CA, 92093-0112 USA. Email: jverstraete@ucsd.edu. Research supported by NSF-BSF award DMS-2347832.

The Erdős–Gallai Theorem [9] asserts that a graph with no path of t edges has at most $(t - 1)n/2$ edges. The Erdős–Sós Conjecture [7] asks for the same bound when the forbidden path is replaced by an arbitrary tree with t edges. This conjecture has received extensive attention in the literature [1–5, 7–11, 16, 17, 20, 22, 23, 25–28, 30]. The Erdős–Sós Conjecture was recently proved employing a permutation argument by GPT-6 Astra AI, described by Adamczewski and Bloom [6, Appendix B.4]. Similar counting arguments are key to the method of Perles [19] giving a very elegant alternative proof of the Erdős–Gallai Theorem using convex geometric graphs, and to extensions to uniform hypergraphs in [14].

Kalai proposed a hypergraph generalization of the Erdős–Sós Conjecture in 1984, recorded by Frankl and Füredi [12], using the following notion of a tree. An r -graph T is a *tight r -tree* if its edges have an ordering $e_1, \dots, e_t$ such that for every $i > 1$, there exists $z_i \in e_i$ and an index $p(i) < i$ satisfying

$$z_i \notin \bigcup_{h < i} e_h, \quad e_i \setminus \{z_i\} \subseteq e_{p(i)}. \quad (2.1)$$

We take $V(T) = \bigcup_{e \in E(T)} e$, so a tight r -tree with t edges has exactly $t + r - 1$ vertices – the case $r = 2$ is precisely a tree in the graph sense. Kalai’s Conjecture is as follows:

Conjecture 1. (Kalai’s Conjecture) *For $n \geq r \geq 2$ and every tight r -tree T with t edges,*

$$\text{ex}(n, T) \leq \frac{t-1}{r} \binom{n}{r-1}. \quad (2.2)$$

To motivate this conjecture, let $b = t + r - 2$ and take a family of b -vertex blocks in which each $(r - 1)$ -set lies in at most one block and almost every $(r - 1)$ -set is covered, and put a complete r -graph on each block. Any copy of a tight tree must stay in a single block – two edges from different blocks cannot meet in $r - 1$ vertices – and therefore each tight tree has at most $b = t + r - 2$ vertices and therefore fewer than t edges. Using Rödl’s packing theorem [24] providing the existence of asymptotically optimal partial Steiner systems, one obtains as $n \rightarrow \infty$:

$$\text{ex}(n, T) \geq \left(\frac{t-1}{r} - o(1) \right) \cdot \binom{n}{r-1}.$$

Keevash’s existence theorem for designs [18] (see also Glock, Kühn, Lo and Osthus [15]) gives exact equality in (2.2) for infinitely many n by covering every $(r - 1)$ -set exactly once with b -sets – these are Steiner systems $S(r - 1, b, n)$. In the special case that the tree T is a sunflower – every edge of T has the form $R \cup \{v\}$ where $|R| = r - 1$ – then any r -uniform hypergraph in which every $(r - 1)$ -set is contained in exactly $t - 1$ hyperedges is T -free. These are sometimes called *simple $(r - 1)$ -($n, r, t - 1$)-designs*.

Several cases of Kalai’s Conjecture were previously known. Frankl and Füredi [12] proved (2.2) for star-shaped tight trees, which have an edge meeting every other edge in $r - 1$ vertices. Füredi, Jiang, Kostochka and the authors [13] proved an asymptotic version for tight trees with a bounded trunk and the exact bound for tight trees with at most four edges. Here a trunk is an initial tight subtree in a construction (2.1) such that each remaining edge attaches directly to an edge of that subtree; the asymptotic statement fixes r and the trunk size and lets t tend to infinity. Stein [27] proved (2.2) for all tight trees when the host r -graph is r -partite.

2.1 Main Theorem

The *shadow* of an r -graph H is

$$\partial H = \{e \subseteq V(H) : |e| = r - 1, e \subseteq h \text{ for some } h \in E(H)\}.$$

We prove Kalai's conjecture by proving the following shadow bound (the equivalence to Kalai's Conjecture was established by Füredi, Jiang, Kostochka and the authors [13, Proposition 2.2]).

Theorem 2.1. *Let $n \geq r \geq 2$, and let T be a tight r -tree with $t \geq 1$ edges. Every n -vertex T -free r -graph H satisfies*

$$|E(H)| \leq \frac{t-1}{r} \cdot |\partial H| \leq \frac{t-1}{r} \cdot \binom{n}{r-1}. \quad (2.3)$$

This theorem also settles Conjecture 1.1 of Füredi, Jiang, Kostochka and the authors [14]. The proof is given in Section 4.

GPT-6 Astra has also give a proof that equality holds in Theorem 2.1 if and only if H is a Steiner $S(r-1, b, n)$ -system ($b = t + r - 2$) or T is a sunflower and every $(r-1)$ -subset of $V(H)$ is contained in exactly $t-1$ edges of H . We have not verified this claim regarding the equality characterization.

2.2 Counting conjecture

The following generalization of the Erdős-Sós Conjecture was stated by the authors in [21]: let T be a tree with t edges. Then there exists $d_0 = d_0(t)$ such that every n -vertex graph G of average degree $d \geq d_0$ contains at least

$$N_T(G) \geq n(d)_t = nd(d-1)(d-2) \cdots (d-t+1) \quad (2.4)$$

labeled copies of T , where $N_T(G)$ denotes the number of injective graph homomorphisms from T to G . The authors in [21] proved the asymptotically sharp estimate

$$N_T(G) \geq \left(1 - O\left(\frac{t^5}{d^2}\right)\right) n(d)_t. \quad (2.5)$$

The conjectured bound (2.4) has subsequently been proved by Wilson [29] for $d_0 = \Omega(t^4)$. A natural strengthening posed by the authors would be to prove the result with

$$d_0(t) = t - 1. \quad (2.6)$$

Such a statement would clearly imply the Erdős-Sós conjecture. It is not too difficult to verify that the conjecture holds for certain small trees, as well as for double stars. A proof, which we have not verified, is claimed by GPT-6 Astra when $d = \Omega(t^3)$ for all trees T and when $d = \Omega(t^2)$ and T is a path with t edges.

We make the following even more general hypergraph counting conjecture:

Conjecture 2. *Let $n \geq r \geq 2$, let $d \geq t - 1 \geq 0$, and let H be an n -vertex r -uniform hypergraph with at least $\frac{d}{r} \binom{n}{r-1}$ edges. Then for any tight tree T with t edges,*

$$N_T(H) \geq (r-1)! \binom{n}{r-1} (d)_t. \quad (2.7)$$

In other words, any hypergraph of average $(r-1)$ -degree $d \geq t-1$ contains at least $(r-1)!\binom{n}{r-1}(d)_t$ labelled copies of any prescribed tight tree T with t edges. This implies Kalai's Conjecture (Conjecture 1), but, as mentioned above, remains open even for paths in graphs. GPT-6 Astra has claimed a proof of Conjecture 2 for $d > 16r^2t^3$. We have not verified this.

2.3 Organization

The special case of trees that are *tight paths* contains the main ideas of the proof of Theorem 2.1, and is presented in Section 3. This proof generalizes the ideas in [14], which previously had the best bound for the extremal number of tight paths. The proof of Theorem 2.1 is given in subsequent sections.

3 Proof of Theorem 2.1 for tight paths

The *tight path* P_k^r has vertex set $\{x_1, \dots, x_{k+r-1}\}$ and edges $\{x_i, \dots, x_{i+r-1}\}$ for $1 \leq i \leq k$. Using convex geometric hypergraphs, Füredi, Jiang, Kostochka and the authors [14, Theorem 1.2] proved

$$\text{ex}(n, P_k^r) \leq \begin{cases} \frac{k-1}{2} \binom{n}{r-1}, & r \text{ even}, \\ \frac{1}{2} \left(k + \left\lfloor \frac{k-1}{r} \right\rfloor \right) \binom{n}{r-1}, & r \text{ odd}. \end{cases} \quad (3.1)$$

The proof of this theorem, which proceeds by considering a count of ordered tight paths in so-called convex geometric hypergraphs [14] partly motivates the proof of Theorem 2.1; indeed in this section we prove the tight path case of Theorem 2.1, as it is much simpler and yet illustrates the main ideas. Roughly speaking, we count permutations whose first $r-1$ vertices, together with a marked later vertex, form an edge of the host hypergraph. For paths, a cyclic rotation extends every eligible state except the first one in each permutation, and these ideas were used to study the extremal function for tight paths in convex geometric hypergraphs [14], and we give this simpler “path argument” proof of Theorem 2.1 in this section. The more complicated case of general trees is achieved in Sections 4.1 – 4.3.

3.1 Definitions and notation

Fix an r -graph H on a vertex set V of size $n \geq r$. For a permutation $\pi = (v_1, \dots, v_n)$ of V , write

$$R(\pi) = (v_1, \dots, v_{r-1}), \quad V_j(\pi) = \{v_1, \dots, v_j\}.$$

Let $\mathcal{W}$ be the set of permutations whose first $r-1$ vertices form a member of ∂H , and define the set of *marked states* by

$$\Omega = \{(\pi, j) : \pi \in \mathcal{W}, r \leq j \leq n, \{v_1, \dots, v_{r-1}, v_j\} \in E(H)\}. \quad (3.2)$$

Set $W = |\mathcal{W}|$ and $M = |\Omega|$. Direct counting gives

$$\begin{aligned} W &= (r-1)! |\partial H| (n-r+1)!, \\ M &= r! |E(H)| (n-r+1)!. \end{aligned} \quad (3.3)$$

For the first identity, choose the initial shadow edge, order it, and order the remaining vertices. For the second, choose an edge, choose its marked vertex in r ways, and order its other $r - 1$ vertices at the beginning of the permutation. Order all remaining $n - r + 1$ vertices arbitrarily; the position of the chosen marked vertex determines j . For $\ell \geq 1$, let $\mathcal{S}_\ell \subseteq \Omega$ consist of the states (π, j) for which $H[V_j(\pi)]$ contains a tight path of ℓ edges ending with the ordered tuple $R(\pi)$. Explicitly, there must be distinct vertices $x_1, \dots, x_{\ell+r-1} \in V_j(\pi)$ such that

$$\text{For } 1 \leq i \leq \ell, \{x_i, \dots, x_{i+r-1}\} \in E(H) \text{ and } (x_{\ell+1}, \dots, x_{\ell+r-1}) = R(\pi). \quad (3.4)$$

The permutation specifies the allowed vertex set and the terminal tuple; the path may traverse its other vertices in any order. Every marked state supports the one-edge path $(v_j, v_1, \dots, v_{r-1})$, so

$$\mathcal{S}_1 = \Omega. \quad (3.5)$$

3.2 Mapping lemma for tight paths

Lemma 3.1. *For every $\ell \geq 1$,*

$$|\mathcal{S}_\ell| \leq |\mathcal{S}_{\ell+1}| + W. \quad (3.6)$$

Proof. For each $\pi \in \mathcal{W}$ having a state in $\mathcal{S}_\ell$, discard (π, j) with j least among its states in $\mathcal{S}_\ell$. At most W states are discarded. We provide an injection Φ' from the remaining states into $\mathcal{S}_{\ell+1}$.

For any $(\pi, j) \in \Omega$, write

$$\pi = (a_1, \dots, a_{r-1}, A, x, B), \quad x = v_j,$$

where $A = (v_r, \dots, v_{j-1})$ and $B = (v_{j+1}, \dots, v_n)$ are possibly empty words. Define

$$\Phi(\pi, j) = (\pi', j), \quad \pi' = (a_2, \dots, a_{r-1}, x, A, a_1, B). \quad (3.7)$$

This rotates the entries in positions $1, \dots, r-1, j$ and fixes all other entries. It preserves the marked edge, and hence maps Ω to itself. Since Φ^r is the identity, Φ is a bijection of Ω . Moreover,

$$V_j(\pi') = V_j(\pi), \quad R(\pi') = (a_2, \dots, a_{r-1}, x). \quad (3.8)$$

Now let $(\pi, j) \in \mathcal{S}_\ell$ be a state that has not been discarded in the first step. There exists $i < j$ with $(\pi, i) \in \mathcal{S}_\ell$. A witnessing path for this earlier state lies in $V_i(\pi)$ and ends with $(a_1, \dots, a_{r-1})$. Since $x = v_j \notin V_i(\pi)$, appending x introduces a new vertex and the new edge $\{a_1, \dots, a_{r-1}, x\} \in E(H)$. The resulting tight path has $\ell + 1$ edges, lies in $V_j(\pi')$, and ends with $R(\pi')$. Thus $\Phi(\pi, j) \in \mathcal{S}_{\ell+1}$. The restriction of the bijection Φ is the required injection Φ' . $\square$

Proof of Theorem 2.1 for tight paths. We aim to show that if H is an n -vertex r -uniform P_k^r -free hypergraph, then $|E(H)| \leq (k-1)/r \cdot |\partial H|$. The assertion is immediate if $E(H) = \emptyset$. Otherwise $n \geq r$, so the preceding definitions apply. If H is P_k^r -free, then $\mathcal{S}_k = \emptyset$. Iterating Lemma 3.1 and using (3.5) gives

$$M = |\mathcal{S}_1| \leq (k-1)W.$$

Substituting (3.3) and cancelling the factor $(r-1)!(n-r+1)!$ yields $r|E(H)| \leq (k-1)|\partial H|$. $\square$

4 Proof of Theorem 2.1

For general tight trees, we prescribe the image of an entire root edge as in Adamczewski and Bloom [6, Appendix B.4]. A prefix permutation inequality then allows two rooted pieces to be joined without an additional error term. This part is not necessary in the proof given in Section 3, as we may repeatedly remove leaf edges and map them to leaf edges of shorter tight paths, whereas for a tight tree, we cannot map leaf edges to leaf edges. Induction then combines this joining operation with the addition of a leaf edge.

We retain the notation H , V , $\mathcal{W}$, Ω , W and M from Section 3. The proof for trees uses embeddings with a prescribed image of a whole root edge. The following straightforward decomposition lemma allows us to do induction on the size of the tree.

Lemma 4.1. *Let T be a tight r -tree with at least two edges, and fix $e \in E(T)$. At least one of the following holds.*

- (i) *T is obtained from a smaller tight tree S by adding a new vertex z and the edge $e = F \cup \{z\}$, where F is an $(r-1)$ -subset of an edge f of S .*
- (ii) *$T = T_1 \cup T_2$ for tight trees T_1, T_2 , each with fewer edges than T , such that*

$$E(T_1) \cap E(T_2) = \{e\}, \quad V(T_1) \cap V(T_2) = e.$$

This lemma is necessary for our inductive proof, which maps leaves to leaves when T is a tight path (as happens in Case (i)), but which may map leaves to non-leaf edges when T is not a path (Case (ii)). We omit the proof of Lemma 4.1, which is an easy exercise.

4.1 States rooted at an edge

We now generalize the definition of $\mathcal{S}_\ell$ in Section 3.1 from tight paths to arbitrary hypergraphs. Let S be an r -graph with a distinguished ordered edge $\vec{e} = (u_1, \dots, u_r)$. For $V(H) = \{v_1, \dots, v_n\}$ and $\pi = v_1 v_2 \dots v_n$, a state $(\pi, j) \in \Omega$ is *good for $(S, \vec{e})$* if there is an injective embedding $\varphi : V(S) \rightarrow V(H)$ such that for $1 \leq i \leq r-1$,

$$\begin{aligned} \varphi(u_i) &= v_i, & \varphi(u_r) &= v_j, \\ \varphi(V(S)) &\subseteq V_j(\pi). \end{aligned} \tag{4.1}$$

Here an embedding sends every edge of S to an edge of H . Let $C(S, \vec{e})$ be the number of good states. Each state is counted once, regardless of how many embeddings witness it.

This number does not depend on the ordering of e . To see this, write $q_i = i$ for $i < r$ and $q_r = j$. If the root is reordered as $(u_{\sigma(1)}, \dots, u_{\sigma(r)})$, permute the host entries in these positions so that

$$v'_{q_i} = v_{q_{\sigma(i)}} \quad (1 \leq i \leq r),$$

and keep all other entries fixed. This is a bijection on Ω , preserves $V_j(\pi)$, and makes the same embedding a witness for the reordered root. We therefore write $C(S, e)$, choosing an ordering of the fixed edge e when convenient. The count can depend on the choice of e . For the one-edge tree,

$$C(e, e) = M. \tag{4.2}$$

4.2 Adding a leaf

The following lemma deals with Case (i) of Lemma 4.1 in the decomposition of a tight tree, and is the analog of the case of paths (Lemma 3.1) in Section 3.

Lemma 4.2. *Let S be an r -graph, let $f \in E(S)$, and let $F \subset f$ have size $r - 1$. Form T by adding a new vertex z and the edge $e = F \cup \{z\}$. Then*

$$C(S, f) \leq C(T, e) + W. \quad (4.3)$$

Proof. Write $F = \{u_1, \dots, u_{r-1}\}$ and $f = F \cup \{y\}$. Choose the root orders $(u_1, \dots, u_{r-1}, y)$ for S and $(u_1, \dots, u_{r-1}, z)$ for T . For every $\pi \in \mathcal{W}$, discard its first good state for (S, f) , if one exists. This discards at most W states.

For a retained state (π, j) , there is an earlier good state (π, i) with $i < j$. Choose its witnessing embedding φ of S . It maps u_h to v_h for $h < r$, maps y to v_i , and has image in $V_i(\pi)$. Since $v_j \notin V_i(\pi)$, extend φ by sending z to v_j . The additional edge maps to $\{v_1, \dots, v_{r-1}, v_j\} \in E(H)$, and the resulting embedding satisfies (4.1) for (T, e) at (π, j) . Thus every retained state is itself good for (T, e) , proving (4.3). $\square$

4.3 Joining along root edges

The more complicated decomposition offered by Case (ii) of Lemma 4.1 is the subject of this section. For a finite set U , let $\text{Perm}(U)$ denote the set of permutations of U . For $w = (w_1, \dots, w_m) \in \text{Perm}(U)$, write

$$w_{\leq s} = (w_1, \dots, w_s), \quad w_{> s} = (w_{s+1}, \dots, w_m).$$

The support of a permutation or one of its blocks is the set of its entries. When $|U| = m$ and $\mathcal{F} \subseteq 2^U$, put

$$C_U(\mathcal{F}) = \{(w, s) : w \in \text{Perm}(U), 0 \leq s \leq m, \text{supp}(w_{\leq s}) \in \mathcal{F}\}$$

and $c_U(\mathcal{F}) = |C_U(\mathcal{F})|$. The empty prefix is included in $C_U(\mathcal{F})$. Thus the total number of pairs (w, s) is $(m+1)m! = (m+1)!$.

In Lemma 4.3 below, we should think of a tight tree T being expressed as the union of two tight trees T_1 and T_2 with a common edge e so $T = T_1 \cup T_2$ and $V(T_1) \cap V(T_2) = e$, and $\mathcal{A}, \mathcal{B}, \mathcal{C}$ as vertex sets of the underlying hypergraph that support T_1, T_2 , and T , respectively, where (the image of) e is excluded.

Lemma 4.3. *Suppose $\mathcal{A}, \mathcal{B}, \mathcal{C} \subseteq 2^U$ satisfy $\mathcal{C} \subseteq \mathcal{A}$ and*

$$R \in \mathcal{A}, \quad X \in \mathcal{B}, \quad R \cap X = \emptyset \implies R \cup X \in \mathcal{C}. \quad (4.4)$$

If $|U| = m$, then

$$c_U(\mathcal{A}) + c_U(\mathcal{B}) \leq (m+1)! + c_U(\mathcal{C}). \quad (4.5)$$

Proof. We construct an injection from the set of pairs $(w, s) \in C_U(\mathcal{A})$ for which $\text{supp}(w_{\leq s}) \notin \mathcal{C}$ to the set of all pairs (w', s') with $\text{supp}(w'_{\leq s'}) \notin \mathcal{B}$. This will imply $c_U(\mathcal{A}) - c_U(\mathcal{C}) \leq (m+1)! - c_U(\mathcal{B})$ as required.

Let (w, s) be a pair of the first kind, and let R be the shortest prefix of w with $\text{supp}(R) \in \mathcal{A} \setminus \mathcal{C}$. This prefix exists and has length at most s . Write

$$w = RXY, \quad s = |R| + |X|, \quad w' = XRY, \quad s' = |X|$$

where juxtaposition denotes concatenation, and define

$$(w, s) \mapsto (w', s'). \quad (4.6)$$

If $\text{supp}(X) \in \mathcal{B}$, then (4.4) would imply $\text{supp}(RX) \in \mathcal{C}$, contrary to the assumption on the input pair. Hence the output has prefix outside $\mathcal{B}$.

Given (w', s') let us now show how to recover (w, s) uniquely, thus showing that this map is injective. First observe that, given w' , the integer s' determines the word X . In the remaining suffix $w'_{>s'} = RY$, take the shortest prefix whose support belongs to $\mathcal{A} \setminus \mathcal{C}$. This prefix is exactly R , since its support belongs to $\mathcal{A} \setminus \mathcal{C}$, and each of its shorter prefixes is the corresponding shorter prefix of the original word w , which does not belong to $\mathcal{A} \setminus \mathcal{C}$. We thereby recover R and Y , and hence (w, s) . This also applies when R or X is empty. $\square$

In the application to tight trees, we have the stronger inclusion $\mathcal{C} \subseteq \mathcal{A} \cap \mathcal{B}$: an embedding of the whole tree restricts to an embedding of either rooted subtree. However, the lemma requires only $\mathcal{C} \subseteq \mathcal{A}$.

Lemma 4.4. *Suppose $T = T_1 \cup T_2$, where*

$$V(T_1) \cap V(T_2) = e, \quad e \in E(T_1) \cap E(T_2).$$

Then

$$C(T_1, e) + C(T_2, e) \leq M + C(T, e). \quad (4.7)$$

Proof. Fix an ordering $e = (u_1, \dots, u_r)$ and an assignment $u_i \mapsto a_i$ to distinct vertices with $\{a_1, \dots, a_r\} \in E(H)$. Put

$$A_0 = \{a_1, \dots, a_r\}, \quad U = V(H) \setminus A_0, \quad m = n - r.$$

A marked state with this root-edge assignment is of the form

$$((a_1, \dots, a_{r-1}, P, a_r, Q), r + |P|).$$

Consequently, for $w = (P, Q)$, these marked states are in bijection with all pairs (w, s) , where $w \in \text{Perm}(U)$ and $0 \leq s \leq m$:

$$(w, s) \longleftrightarrow ((a_1, \dots, a_{r-1}, w_{\leq s}, a_r, w_{>s}), r + s). \quad (4.8)$$

There are $(m + 1)!$ such states. In particular, $s = 0$ is allowed: the empty prefix outside A_0 corresponds to a marked state.

Let $\mathcal{A}$, $\mathcal{B}$ and $\mathcal{C}$ be the families of sets $X \subseteq U$ for which T_1 , T_2 and T , respectively, embed into $H[A_0 \cup X]$ with the prescribed assignment on e . Clearly $\mathcal{C} \subseteq \mathcal{A}$, since if we restrict an embedding of $\mathcal{C}$ we get an embedding of $\mathcal{A}$. If $R \in \mathcal{A}$, $X \in \mathcal{B}$ and $R \cap X = \emptyset$, choose the two witnessing embeddings. They agree on every vertex of e and have disjoint images outside e . Since $V(T_1) \cap V(T_2) = e$, their

union is an injective embedding of T into $H[A_0 \cup R \cup X]$. Hence $R \cup X \in \mathcal{C}$. Consequently, $\mathcal{A}, \mathcal{B}, \mathcal{C}$ satisfy the hypothesis of Lemma 4.3.

We now sum the conclusion of Lemma 4.3 to obtain the conclusion of Lemma 4.4. For clarity of presentation, we make the counting explicit. Keep the ordering $e = (u_1, \dots, u_r)$ fixed, and let $\mathcal{I}_H$ be the set of ordered edges of H , so that $|\mathcal{I}_H| = r!|E(H)|$. For each $\mathbf{a} = (a_1, \dots, a_r) \in \mathcal{I}_H$, define

$$\Omega_{\mathbf{a}} = \{(\pi, j) \in \Omega : v_i = a_i \text{ for } 1 \leq i < r, \quad v_j = a_r\}.$$

Let $C_{\mathbf{a}}(S, e)$ be the number of states in $\Omega_{\mathbf{a}}$ that are good for (S, e) . Every state $(\pi, j) \in \Omega$ determines the unique assignment $\mathbf{a} = (v_1, \dots, v_{r-1}, v_j)$. Consequently, the sets $\Omega_{\mathbf{a}}$ partition Ω , and, for every S under consideration,

$$C(S, e) = \sum_{\mathbf{a} \in \mathcal{I}_H} C_{\mathbf{a}}(S, e). \quad (4.9)$$

We count each good state once, regardless of how many embeddings satisfy its defining conditions. Now fix $\mathbf{a} \in \mathcal{I}_H$, and define A_0 , U and the families $\mathcal{A}, \mathcal{B}, \mathcal{C}$ as above. Under the bijection (4.8), a pair (w, s) corresponds to the state

$$\pi = (a_1, \dots, a_{r-1}, w_1, \dots, w_s, a_r, w_{s+1}, \dots, w_m), \quad j = r + s.$$

Its first j vertices form exactly the set

$$V_j(\pi) = A_0 \cup \{w_1, \dots, w_s\}.$$

Moreover, the prescribed images of the root vertices are $u_i \mapsto a_i$ for every $1 \leq i \leq r$. It follows that this state is good for (T_1, e) if and only if $\{w_1, \dots, w_s\} \in \mathcal{A}$. The corresponding equivalences hold for (T_2, e) and $\mathcal{B}$, and for (T, e) and $\mathcal{C}$. Therefore

$$C_{\mathbf{a}}(T_1, e) = c_U(\mathcal{A}), \quad C_{\mathbf{a}}(T_2, e) = c_U(\mathcal{B}), \quad C_{\mathbf{a}}(T, e) = c_U(\mathcal{C}).$$

Lemma 4.3 consequently gives, for this assignment,

$$C_{\mathbf{a}}(T_1, e) + C_{\mathbf{a}}(T_2, e) \leq (m+1)! + C_{\mathbf{a}}(T, e).$$

Finally, $m = n - r$ is the same for every assignment. Summing the preceding inequality over $\mathcal{I}_H$ and using (4.9), we obtain

$$\begin{aligned} C(T_1, e) + C(T_2, e) &= \sum_{\mathbf{a} \in \mathcal{I}_H} (C_{\mathbf{a}}(T_1, e) + C_{\mathbf{a}}(T_2, e)) \\ &\leq \sum_{\mathbf{a} \in \mathcal{I}_H} ((m+1)! + C_{\mathbf{a}}(T, e)) \\ &= r!|E(H)|(m+1)! + C(T, e) \\ &= M + C(T, e), \end{aligned}$$

where the last equality follows from $m+1 = n - r + 1$ and (3.3). This proves (4.7). $\square$

4.4 Completing the proof of Theorem 2.1

In this section we obtain an upper bound for M by applying Lemmas 4.2 and 4.4, and use this to complete the proof of Theorem 2.1.

Theorem 4.5. *For every tight r -tree T with $t \geq 1$ edges and every $e \in E(T)$,*

$$M \leq C(T, e) + (t - 1)W. \quad (4.10)$$

Proof. We proceed by induction on t . For $t = 1$, the claim is (4.2). Suppose $t \geq 2$, and apply Lemma 4.1 at e . In case (i), the smaller tight tree S has $t - 1$ edges and for some edge $f \in E(S)$ the edge e intersects f in exactly $r - 1$ vertices. The induction hypothesis rooted at f , followed by Lemma 4.2, gives

$$M \leq C(S, f) + (t - 2)W \leq C(T, e) + (t - 1)W.$$

In case (ii), write $t_i = |E(T_i)|$. Then $t_i < t$ for $i = 1, 2$ and $t_1 + t_2 = t + 1$. Add the two induction inequalities and apply Lemma 4.4 to obtain

$$\begin{aligned} 2M &\leq C(T_1, e) + C(T_2, e) + (t_1 + t_2 - 2)W \\ &\leq M + C(T, e) + (t - 1)W. \end{aligned}$$

Subtracting M completes the induction. $\square$

Proof of Theorem 2.1. If $E(H) = \emptyset$, the assertion is immediate. Otherwise $n \geq r$. Fix any edge e of T . Since H is T -free, $C(T, e) = 0$. Theorem 4.5 and (3.3) give

$$r! |E(H)| (n - r + 1)! = M \leq (t - 1)W = (t - 1)(r - 1)! |\partial H| (n - r + 1)!.$$

Cancelling $(r - 1)!(n - r + 1)!$ proves $r|E(H)| \leq (t - 1)|\partial H|$. Finally, $|\partial H| \leq \binom{n}{r-1}$ gives (2.2). $\square$

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